

Francisco Javier Polanco-Rodríguez

Postdoctoral Researcher (ATER)

Status

Nationality Spanish
Languages Spanish (native), English (fluent), French (fluent)
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Professional Experience

2026–2027 **Postdoctoral researcher (ATER)**, *École Polytechnique, Laboratoire de Physique des Plasmas (LPP), Palaiseau (France)*, October 2026 – March 2027

Radio emission mechanisms of type III solar radio bursts in the solar wind: PIC simulations, theory and Solar Orbiter observations (space plasmas group).

2023–2026 **PhD researcher in Physics**, *Université Paris-Saclay, Laboratoire de Physique des Plasmas (LPP), France*, PhD defended on September 18, 2026, Palaiseau

Electromagnetic Emissions at the Plasma Frequency by Type III Solar Radio Bursts in Solar Wind and Coronal Plasmas

Doctoral School n°572 Ondes et Matière (EDOM)

Advisors: Catherine Krafft (Université Paris-Saclay) & Philippe Savoini (Sorbonne Université)

Jury
President K. Bocchialini (Université Paris-Saclay, IAS, Orsay)
Referees T. Dudok de Wit (Université d'Orléans, LPC2E, Orléans)
P. Henri (CNRS, Lab. Lagrange, Nice)
Examiners F. Califano (University of Pisa, Italy)
M. Maksimovic (CNRS, LIRA, Meudon)
C. Riconda (Sorbonne Université, LULI, Palaiseau)
N. Vilmer (CNRS, LIRA, Meudon)

Education

2022–2023 **Master 2 Plasma Physics and Fusion**, *Institut Polytechnique de Paris (France)*

Average of 17.1/20, class rank: 2/20.

Awarded the excellence scholarship of the EUR PlasmaScience.

Master 2 internship (2023): Polarization characteristics of electromagnetic emissions by type III solar radio bursts.

Laboratoire de Physique des Plasmas (LPP)

2018–2022 **Bachelor in Fundamental Physics**, *University of Seville (Spain)*

Average of 8.65/10.

Research Activity

Topics *Plasma physics – Solar radio bursts – Plasma emission – Solar wind and corona – Radio waves – High-performance Particle-in-Cell simulations – Spacecraft data analysis.*

Publications **6 publications** in peer-reviewed journals: 4 ApJL, 1 Nature Astronomy and 1 A&A; 4 as first author. **1 publication** submitted to ApJL. **1 review** (180 pages) to be submitted in 2026 to Space Science Reviews. **2 publications** in preparation (to be submitted to ApJL), 1 as first author.

Communications 14 oral presentations at international conferences (10 as speaker), including 2 seminars.
5 poster presentations at international conferences (including AGU Fall Meeting 2026).

Teaching Experience

2026–2027 **48 hours as ATER at *Université Paris-Saclay*** (upcoming)

- 48h of numerical simulation courses for Master 2 students.

2023–2026 **192 hours at *Université Paris-Saclay***

- 148h of practical laboratory sessions in plasma physics. Students of the Master 1 "Physique Fondamentale" and the "Magistère 2 de Physique" at *Université Paris-Saclay*: 8h sessions with 12 students working in pairs. Students of the "Double Diplôme" Master 1 from engineering schools (e.g. ENS Paris-Saclay, Institut d'Optique Graduate School, CentraleSupélec): 4h sessions with 6 students working in pairs (18:00–22:00).

- 44h of lectures and practical courses in scientific Python programming for first-year bachelor's students (Mathematics and Physics): 4h sessions combining lectures and hands-on exercises, with 20 students.

2023–2025 **24 hours at the Summer School on Plasma Physics (Plas@PAR), Roscoff**
Lectures on test-particle simulations in a Python/ParaView environment (3h sessions, 26 students).

Outreach

2022–2026 **Fête de la Science (Science Festival), *École Polytechnique***

Participation in the Science Festival of *École Polytechnique* in 2022, 2023, 2025 and 2026. Presentations and demonstrations of plasma physics for audiences aged 7 to 70, over 2 days each year.

List of Peer-Reviewed Publications

[P1] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2026, Astrophys. J. Lett., 1002, L35, *Evidence of Langmuir/$\mathcal{Z}$-mode Wave Decay into $\mathcal{Z}$-mode Electromagnetic Radiation in the Solar Wind*

[P2] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2026, Astron. & Astrophys., 711, A139, *Analysis of wave processes using beam-driven Langmuir/$\mathcal{Z}$-mode waveforms generated in Particle-in-Cell simulations*

[P3] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2025, Astrophys. J. Lett., 982, L24, *Decay of Turbulent Upper-hybrid Waves in Weakly Magnetized Solar Wind Plasmas*

[P4] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2025, Astrophys. J. Lett., 989, L38, *Polarization Ratios of Turbulent Langmuir/Z-mode Waves Generated by Electron Beams in Magnetized Solar Wind Plasmas*

[P5] Krafft, C., **Polanco-Rodríguez**, F.J. & Savoini, P., 2026, submitted to Astrophys. J. Lett., *Evidence of linear mode conversion of Langmuir/$\mathcal{Z}$-mode waves to electromagnetic waves in the solar wind*

[P6] Krafft, C., Volokitin, A.S., **Polanco-Rodríguez**, F.J. & Savoini, P., 2025, Nat. Astron., 9, 1292, *Radiation efficiency of electromagnetic wave modes from beam-generated solar radio sources*

[P7] Krafft, C., Savoini, P. & **Polanco-Rodríguez**, F.J., 2024, Astrophys. J. Lett., 967, L20, *Mechanisms of Fundamental Electromagnetic Wave Radiation in the Solar Wind*

In preparation / to be submitted

[P8] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., in preparation for Astrophys. J. Lett., 2026, *Random Density Fluctuations at Sub-ion Scales in the Solar Wind*

[P9] Krafft, C., **Polanco-Rodríguez**, F.J. & Savoini, P., in preparation for Astrophys. J. Lett., 2026, *Electromagnetic decay of Langmuir/Z-mode waves in solar wind plasmas*

[P10] Krasnoselskikh, V., Krafft, C. et al., 2026, review to be submitted to Space Science Reviews in 2026 (180 pages, 15 authors), *Beam-Plasma Interactions and the Generation of Type III Radio Bursts*

List of Scientific Communications

Oral presentations *(the speaker name is underlined)*

[OP1] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2026, RPW science meeting, Nijmegen, Netherlands (online), *Evidence of Langmuir/Z-mode wave decay into electromagnetic Z-mode waves in the solar wind*

[OP2] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2026, RPW science meeting, Kiruna, Sweden, *Evidence of Langmuir/Z-mode wave decay into electromagnetic Z-mode waves in the solar wind*

[OP3] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2026, Astrophysics Day, Université Paris-Saclay, Orsay, France, *Radio efficiency of electromagnetic wave modes from beam-generated solar radio sources*

[OP4] **Polanco-Rodríguez**, F.J., Krafft, C., Savoini, P. & Volokitin, A.S., 2026, seminar at LPC2E, Orléans, France, *Radio emission by electron beams during type III solar bursts*

[OP5] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2026, ATST conference, Biarritz, France, *Evidence of Langmuir/Z-mode Wave Decay into Z-mode Electromagnetic Radiation in the Solar Wind*

[OP6] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2026, LPP space plasmas group seminar, Palaiseau, France, *Evidence of Langmuir/Z-mode wave decay into electromagnetic Z-mode waves in the solar wind*

[OP7] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2025, 5th SMILEI user & training workshop, Madrid, Spain, *Wave processes during type III solar radio bursts: 2D PIC simulations*

[OP8] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2025, In situ heliospheric science meeting, Écully, France, *Electromagnetic Z-mode wave radiation in solar wind plasmas*

[OP9] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2024, 4th SMILEI user & training workshop, Prague, Czech Republic, *PIC simulations of electromagnetic emissions by solar radio bursts*

[OP10] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2024, Elbereth conference, Paris, France, *Study of nonlinear processes occurring during type III solar radio bursts using waveform analysis*

[OP11] Krafft, C., Volokitin, A.S., **Polanco-Rodríguez**, F.J. & Savoini, P., 2025, International Workshop on Planetary, Solar and Heliospheric Radio Emissions (PRE), Marseille, France, *Electromagnetic wave radiation in turbulent magnetized and inhomogeneous plasmas*

[OP12] Krafft, C., Volokitin, A.S., **Polanco-Rodríguez**, F.J. & Savoini, P., 2025, EGU General Assembly, Vienna, Austria, *Electromagnetic wave radiation in turbulent magnetized and inhomogeneous plasmas: theory and simulations*

[OP13] Krafft, C., Savoini, P., **Polanco-Rodríguez**, F.J. & Volokitin, A.S., 2024, 50th EPS Conference on Plasma Physics, Salamanca, Spain, *Mechanisms of fundamental electromagnetic wave radiation in the solar wind*

[OP14] Krafft, C., Savoini, P. & **Polanco-Rodríguez**, F.J., 2024, Joint SolO/RPW & PSP/FIELDS workshop, Orléans, France, *Wave interaction and conversion processes revealed by Langmuir waveforms in 2D PIC numerical simulations*

Poster presentations

[PP1] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2026 (December, upcoming), AGU Fall Meeting, San Francisco, USA, *Mechanisms of Radio Emission at the Plasma Frequency During Type III Solar Radio Bursts: PIC Simulations, Theory, and Solar Orbiter Observations*

[PP2] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2025, 5th SMILEI user & training workshop, Madrid, Spain, *Wave processes during type III solar radio bursts: 2D PIC simulations*

[PP3] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2025, EGU General Assembly, Vienna, Austria, *Wave processes during type III solar radio bursts: 2D PIC simulations*

[PP4] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2024, PNST, Marseille, France, *2D PIC simulations of type III solar radio bursts: a virtual satellite approach study*

[PP5] **Polanco-Rodríguez**, F.J., Krafft, C. & Savoini, P., 2024, 50th EPS Conference on Plasma Physics, Salamanca, Spain, *Study of wave processes occurring during type III solar radio bursts through PIC simulations*